

Productivity, Output, and Employment

ECON202 Macroeconomics — UG-I, 2025: Semester III

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Sections Covered

§	Topic
3-1	The Production Function
3-2	The Demand for Labor
3-3	The Supply of Labor
3-4	Labor Market Equilibrium
3-5	Unemployment
3-6	Okun's Law

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Learning Objectives

1. Discuss production function properties and changes
2. Discuss factors that affect the demand for labor
3. Discuss the factors that affect the supply of labor
4. Identify factors that affect labor market equilibrium
5. Explain how the unemployment rate is measured and describe changes in employment status
6. Explain the significance of Okun's law

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► 3-1 · The Production Function

3-2 · The Demand for Labor

3-3 · The Supply of Labor

3-4 · Labor Market Equilibrium

3-5 · Unemployment

3-6 · Okun's Law

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The Production Function - Introduction

| 3-1 · THE PRODUCTION FUNCTION

- **Factors of production**
 - Capital (K)
 - Labor (N)
 - Others (raw materials, land, energy)
 - Productivity of factors depends on technology and management
- **General form**

$$Y = AF(K, N) \quad (3.1)$$

- Parameter A is **total factor productivity** — the effectiveness with which capital and labor are used

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The Production Function - Cobb–Douglas Form

| 3-1 · THE PRODUCTION FUNCTION

- **Cobb–Douglas specification**

$$Y = AK^\alpha N^\beta \quad (3.2)$$

- α, β : output elasticities of capital and labor
- TFP cannot be measured directly, but it can be measured indirectly, as a residual. Also known as the **Solow residual**

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Constant Returns to Scale & Production Elasticities

| 3-1 · THE PRODUCTION FUNCTION

- **Constant Returns to Scale:** When $\alpha + \beta = 1$
 - If you double all inputs, output exactly doubles
 - Foundation of neoclassical growth theory
- **Production Elasticities:**
 - α = elasticity of output with respect to capital
 - β = elasticity of output with respect to labor

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Income Shares: The Dual Interpretation

| 3-1 · THE PRODUCTION FUNCTION

- In competitive markets, factors are paid their marginal products
 - Wage rate = Marginal product of labor (MPN)
 - Rental rate = Marginal product of capital (MPK)
- Income Shares:
 - Labor's share of total income = β
 - Capital's share of total income = α
 - Total shares: $\alpha + \beta = 1$
- **Amazing Result:** Production elasticities = Income shares!

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The Production Function - EU Economy Application

| 3-1 · THE PRODUCTION FUNCTION

Cobb-Douglas production function works well for EU economy:

$$Y = A \cdot K^{0.33} \cdot N^{0.67}$$

TABLE 3.1

The Production Function of the EU-28, 2005–2014

Year	(1) Real GDP (Y) in 2010 prices Million EUR	(2) Real Capital Stock (K) in 2010 prices Million EUR	(3) Labor, N (millions of workers)	(4) $A = Y/(K^{0.33}N^{0.67})$	(5) Growth in A (% change in A)
2005	12,230	2660	208	25.35	—
2006	12,647	2848	212	25.32	0.12%
2007	13,037	3031	216	25.55	0.91%
2008	13,103	2990	218	25.38	-0.74%
2009	12,527	2478	214	26.1	2.92%
2010	12,789	2613	212	26.33	0.88%
2011	13,011	2696	212	26.51	0.68%
2012	12,943	2539	211	26.99	1.8%
2013	12,956	2513	210	27.19	0.74%
2014	13,127	2576	213	27.08	0.40%

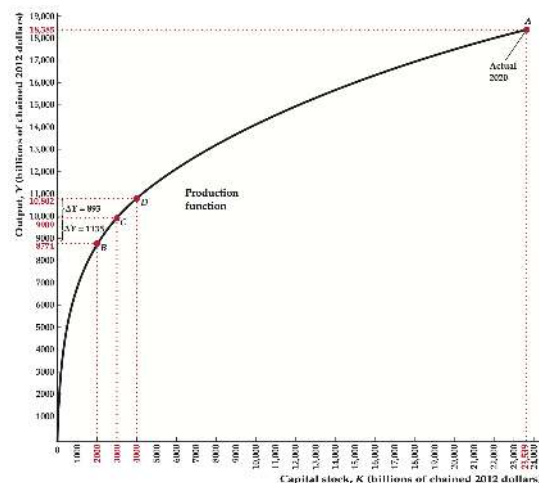
Source: Real GDP, capital stock, and labor force figures are from the Eurostat Database website, <http://ec.europa.eu/eurostat/data/database>.

The Production Function of the EU, 2005–2014

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The Properties of the Production Function

| 3-1 · THE PRODUCTION FUNCTION

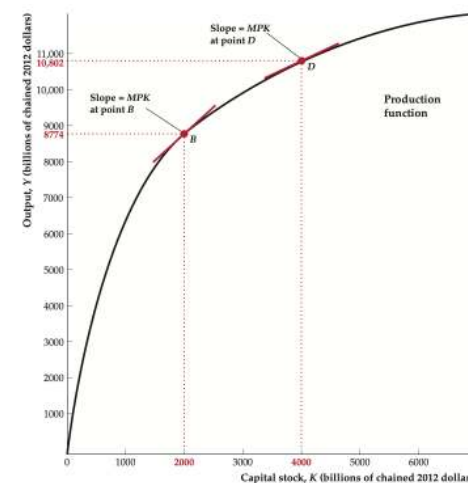


- Slopes upward: more of any input produces more output
- Slope becomes flatter as input rises: diminishing marginal product as input increases

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Marginal Product of Capital: $MPK = \Delta Y / \Delta K$

| 3-1 · THE PRODUCTION FUNCTION

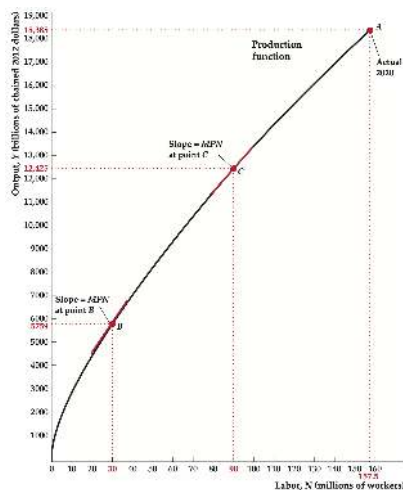


- Equal to slope of production function graph (Y vs K)
- MPK always positive
- Diminishing marginal productivity of capital

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Marginal Product of Labor: $MPN = \Delta Y / \Delta N$

| 3-1 · THE PRODUCTION FUNCTION



- Equal to slope of production function graph (Y vs N)
- MPN always positive
- Diminishing marginal productivity of labor

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Supply Shocks

| 3-1 · THE PRODUCTION FUNCTION

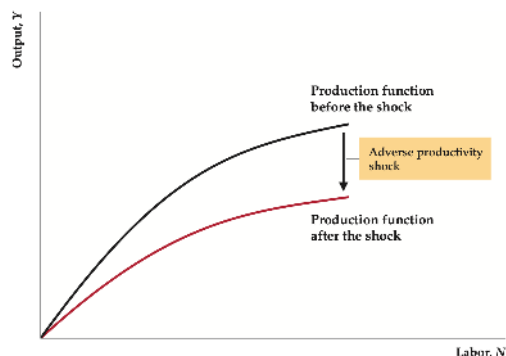
Productivity shock = a change in an economy's production function

- Supply shocks affect the amount of output that can be produced for a given amount of inputs
- Shocks may be positive (increasing output) or negative (decreasing output)
- Examples: weather, inventions and innovations, government regulations, oil prices

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An Adverse Supply Shock that Lowers the MPN

| 3-1 · THE PRODUCTION FUNCTION



- Negative (adverse) shock: Usually slope of production function decreases at each level of input
- Positive shock: Usually slope of production function increases at each level of input (e.g., if parameter **A** increases)

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✓ 3-1 · The Production Function

▶ 3-2 · The Demand for Labor

3-3 · The Supply of Labor

3-4 · Labor Market Equilibrium

3-5 · Unemployment

3-6 · Okun's Law

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The Demand for Labor - Assumptions

| 3-2 · THE DEMAND FOR LABOR

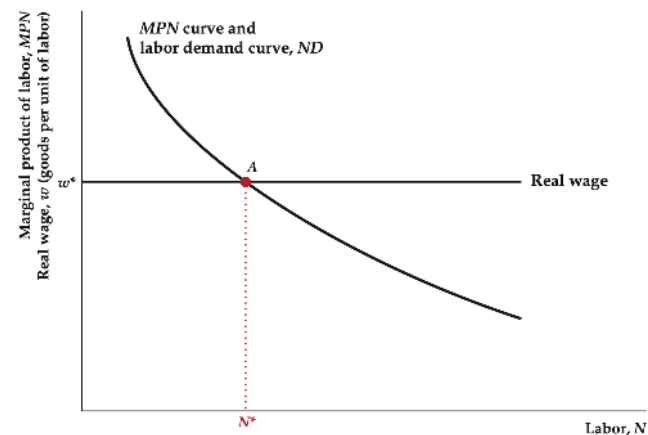
- How much labor do firms want to use?
 - Hold capital stock fixed—short-run analysis
 - Workers are all alike
 - Labor market is competitive
 - Firms maximize profits

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Marginal Analysis for Labor Demand

| 3-2 · THE DEMAND FOR LABOR

Analysis at the margin: costs and benefits of hiring one extra worker



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The Clip Joint Example

| 3-2 · THE DEMAND FOR LABOR

- Setting the nominal wage (W) equal to the marginal revenue product of labor

$$W = MRPN = P \times MPN \quad (3.3)$$

- $W = MRPN$ is the same condition as $w = MPN$, since
 - $W = P \times w$ and
 - $MRPN = P \times MPN$

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The Clip Joint's Production Function, $W = \$240/\text{Day}$.

| 3-2 · THE DEMAND FOR LABOR

Workers, N	Dogs groomed, Y	Marginal product, MPN	$MRPN = MPN \times P$ ($P = \$30$)
0	0	—	—
1	11	11	\$330
2	20	9	\$270
3	27	7	\$210
4	32	5	\$150
5	35	3	\$90
6	36	1	\$30

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The MP of Labor and Labor Demand: Example

| 3-2 · THE DEMAND FOR LABOR

A change in the wage

- Begin at equilibrium where $W = \text{MRPN}$
- A rise in the wage rate means $W > \text{MRPN}$, unless N is reduced so the MRPN rises
- A decline in the wage rate means $W < \text{MRPN}$, unless N rises so the MRPN falls

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Labor Demand Curve Properties

| 3-2 · THE DEMAND FOR LABOR

The marginal product of labor and the labor demand curve

- Labor demand curve shows relationship between the real wage rate and the quantity of labor demanded
- It is the same as the MPN curve, since $w = \text{MPN}$ at equilibrium
- So the labor demand curve is downward sloping; firms want to hire less labor, the higher the real wage

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Factors That Shift Labor Demand -1

| 3-2 · THE DEMAND FOR LABOR

- A change in the wage causes a movement along the labor demand curve, not a shift of the curve
- Supply shocks: a beneficial shock raises $\text{MPN} \rightarrow$ labor demand shifts right (adverse shifts left)
- Capital stock: more capital raises $\text{MPN} \rightarrow$ labor demand shifts right (less capital shifts left)

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Production Function after a +ve Productivity Shock

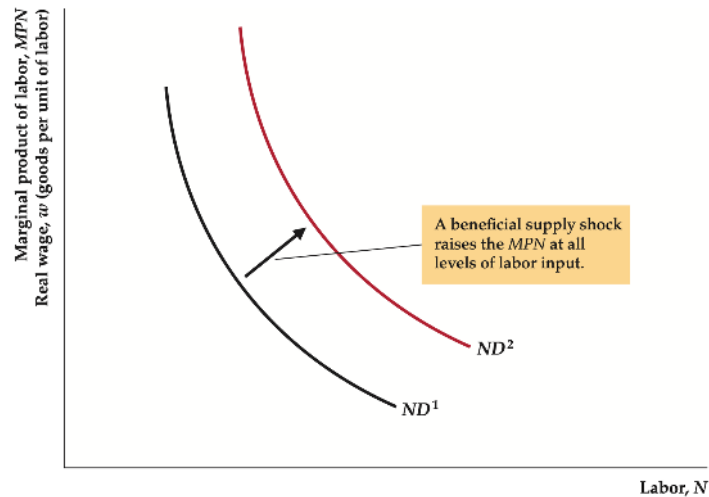
| 3-2 · THE DEMAND FOR LABOR

Workers, N	Dogs groomed, Y	Marginal product, MPN	$\text{MRPN} = \text{MPN} \times P$ ($P = \$30$)
0	0	—	—
1	22	22	\$660
2	40	18	\$540
3	54	14	\$420
4	64	10	\$300
5	70	6	\$180
6	72	2	\$60

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The Effect of a +ve Supply Shock on Labor Demand

| 3-2 · THE DEMAND FOR LABOR



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Aggregate Labor Demand

| 3-2 · THE DEMAND FOR LABOR

- Aggregate labor demand is the sum of all firms' labor demand
- Same factors (supply shocks, size of capital stock) that shift firms' labor demand cause shifts in aggregate labor demand

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Factors That Shift Labor Demand -2

| 3-2 · THE DEMAND FOR LABOR

An increase in	Causes the labor demand curve to shift	Reason
Productivity	Right	Beneficial supply shock increases MPN and shifts MPN curve up and to the right.
Capital stock	Right	Higher capital stock increases MPN and shifts MPN curve up and to the right.

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✓ 3-1 · The Production Function

✓ 3-2 · The Demand for Labor

► **3-3 · The Supply of Labor**

3-4 · Labor Market Equilibrium

3-5 · Unemployment

3-6 · Okun's Law

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Labor Supply Overview

| 3-3 · THE SUPPLY OF LABOR

- Supply of labor is determined by individuals
- Aggregate supply of labor is the sum of individuals' labor supply
- Labor supply of individuals depends on labor-leisure choice



Lunch atop a Skyscraper in New York: 20, 1932

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The Income-Leisure Trade-off

| 3-3 · THE SUPPLY OF LABOR

- Utility depends on consumption and leisure
- Need to compare costs and benefits of working another day
 - Costs: Loss of leisure time
 - Benefits: More consumption, since income is higher
- If benefits of working another day > costs, work another day
- Keep working additional days **until** benefits = costs



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Income and Substitution Effects

| 3-3 · THE SUPPLY OF LABOR

An increase in the real wage has offsetting income and substitution effects

- Substitution effect: a higher wage raises the reward for working → work more
- Income effect: a higher wage makes you richer at the same hours → afford more leisure, work less

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Pure Effects

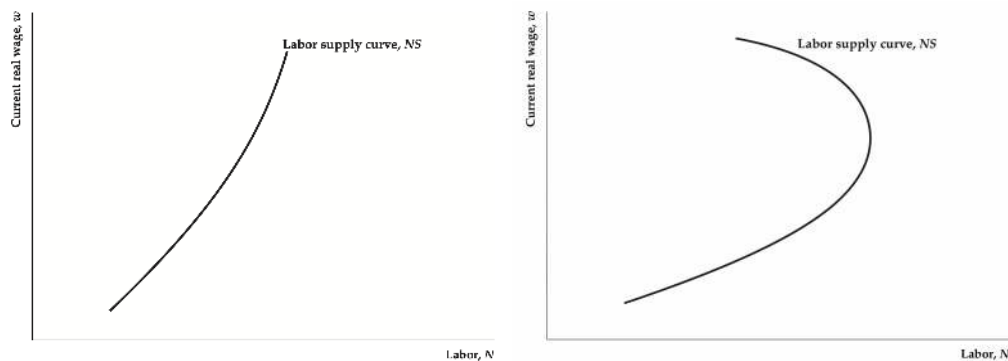
| 3-3 · THE SUPPLY OF LABOR

- A pure substitution effect: a one-day rise in the real wage
 - A temporary real wage increase has just a pure substitution effect, since the effect on wealth is negligible
- A pure income effect: winning the lottery
 - Winning the lottery doesn't have a substitution effect, because it doesn't affect the reward for working
 - But winning the lottery makes a person wealthier, so a person will both consume more goods and take more leisure; this is a pure income effect

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The Labor Supply Curve of an Individual Worker

| 3-3 · THE SUPPLY OF LABOR



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Empirical Evidence: On Real Wages and Labor Supply

| 3-3 · THE SUPPLY OF LABOR

- Overall result: Labor supply increases with a temporary rise in the real wage
- Labor supply falls with a permanent increase in the real wage

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Factors That Shift Labor Supply

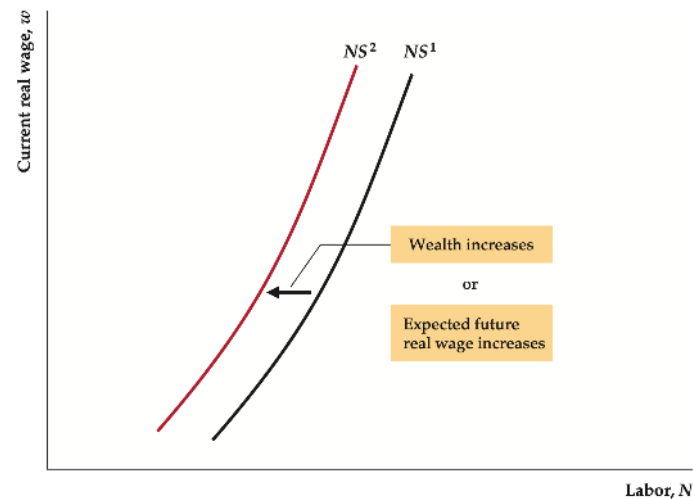
| 3-3 · THE SUPPLY OF LABOR

- Wealth: more wealth reduces labor supply → curve shifts left
- Expected future real wage: a higher expected wage acts like more wealth → curve shifts left

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The Effect on Labor Supply of an Increase in Wealth

| 3-3 · THE SUPPLY OF LABOR



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Why Does the Aggregate Labor Supply Curve Slope Upward?

| 3-3 · THE SUPPLY OF LABOR

Aggregate labor supply rises when current real wage rises

- Some people work more hours. **Intensive Margin**
- Other people enter labor force **Extensive Margin**

How elastic is the LS? **Imp. Macro Question:** (Keane & Rogerson, 2012)

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Factors Increasing Labor Supply

| 3-3 · THE SUPPLY OF LABOR

- Decrease in wealth
- Decrease in expected future real wage
- Increase in working-age population (higher birth rate, immigration)
- Increase in labor force participation (increased female labor participation, elimination of mandatory retirement)

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Why LS Matters for Macroeconomics

| 3-3 · THE SUPPLY OF LABOR

(Rogerson, 2024) suggest an important role for labor supply in affecting macroeconomic outcomes

- Large differences in hours of work across OECD¹ economies imply large differences in GDP per capita
- Large differences in the size of tax and transfer programs across countries (proxied by government revenues relative to GDP)
- These two outcomes are strongly negatively correlated

1. Organisation for Economic Co-operation and Development

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Recent Labor Supply Trends

| 3-3 · THE SUPPLY OF LABOR

- Participation has fallen since 2010 as baby boomers retire
- Prime-age (25–54) participation is down too, even for women
- The 2020 pandemic recession pushed participation lower still
- Causes: post-recession discouragement, more disability, the opioid crisis, weak family benefits

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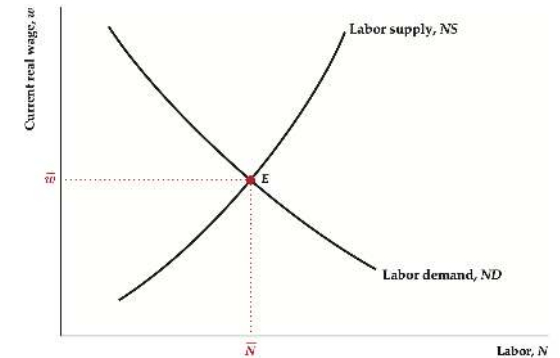
- ✓ 3-1 · The Production Function
- ✓ 3-2 · The Demand for Labor
- ✓ 3-3 · The Supply of Labor
- ▶ **3-4 · Labor Market Equilibrium**
- 3-5 · Unemployment
- 3-6 · Okun's Law

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Labor Market Equilibrium

| 3-4 · LABOR MARKET EQUILIBRIUM

- Classical model of the labor market—real wage adjusts quickly
- Pins down full-employment employment and the market-clearing real wage
- Problem with classical model: can't study unemployment



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Full-Employment Output

| 3-4 · LABOR MARKET EQUILIBRIUM

- Full-employment output = potential output = level of output when labor market is in equilibrium

$$\bar{Y} = AF(K, \bar{N}) \quad (3.4)$$

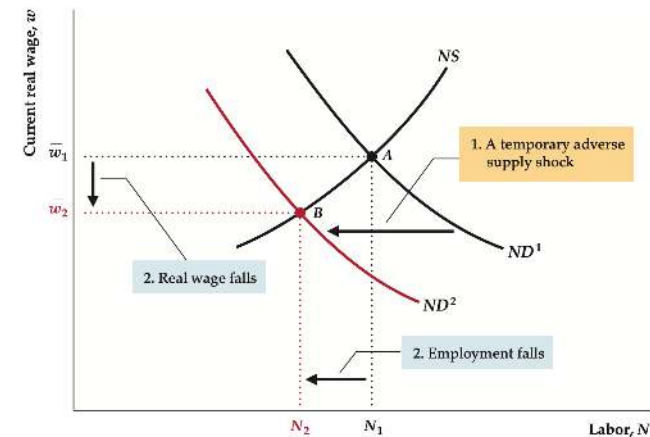
- Affected by changes in full employment level or production function (example: supply shock)

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LM Effects of a Temporary Adverse Supply Shock

| 3-4 · LABOR MARKET EQUILIBRIUM

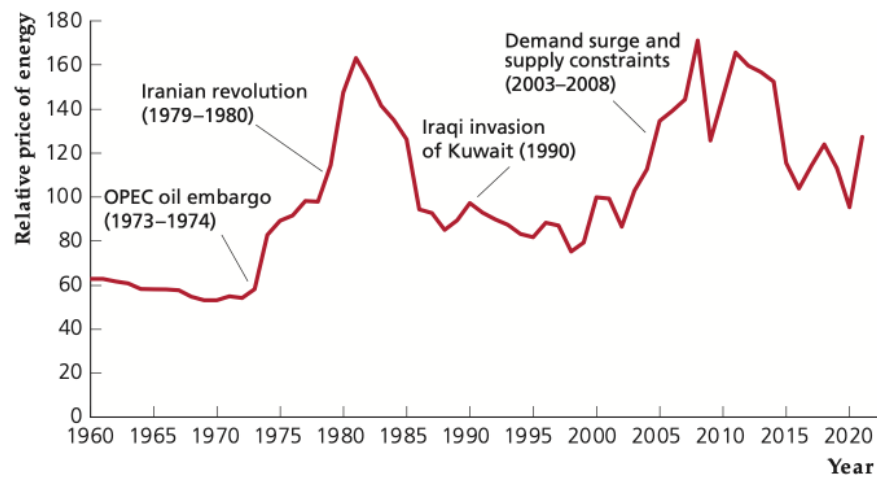
Temporary Adverse Supply Shock?



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Relative Price of Energy, 1960–2021

| 3-4 · LABOR MARKET EQUILIBRIUM



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- ✓ 3-1 · The Production Function
- ✓ 3-2 · The Demand for Labor
- ✓ 3-3 · The Supply of Labor
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- 3-6 · Okun's Law

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Unemployment

| 3-5 · UNEMPLOYMENT

REALITY



UNEMPLOYED

LinkedIn



AI Enthusiast |
GenAI Evangelist |
Influencer | Inspirer |
Chief Visionary | Coach |
Startup Advisor | Mentor |
Prompt Engineer | Project
Manager | CxO |
LinkedIn Top voice
(according to me) |
Speaker | ChatGPT Expert |
9 x AWS Certifications |
7 x Microsoft Certifications |
Father and son

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Why Do We Care About Unemployment?

| 3-5 · UNEMPLOYMENT

- High or low unemployment is a symptom of inefficiency
- High unemployment: many resources are idle
- Low unemployment: labor shortages waste resources on recruiting
- It cuts the welfare of the unemployed, especially in long spells
- **Worse mental and physical health — anxiety, low self-esteem — even controlling for income**

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Employment / Unemployment

| 3-5 · UNEMPLOYMENT

- A person is employed if she has a full-time job or does temporary or part-time work during the survey week
- A person is unemployed if:
 - She does not have a job
 - She is available to work
 - She has been actively looking for a job in the last four weeks (contacting firms, sending CVs, applying to vacancies)

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Measuring Unemployment

| 3-5 · UNEMPLOYMENT

- Categories: employed, unemployed, not in the labor force
- Labor Force = Employed + Unemployed
- Unemployment Rate = Unemployed / Labor Force
- Participation Rate = Labor Force / Adult (working-age) Population
- Employment Ratio = Employed / Adult (working-age) Population
- Discouraged Workers

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Discouraged Workers

| 3-5 · UNEMPLOYMENT

- Discouraged workers are those who give up looking for a job
- They are no longer counted as unemployed
- Some measures of unemployment account for them
- Six measures: U1 – U6
- Official measure: U3

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Employment Status of the U.S. Adult (working-age) Population

| 3-5 · UNEMPLOYMENT

TABLE 3.4

Employment Status of the U.S. Adult Population, January 2022

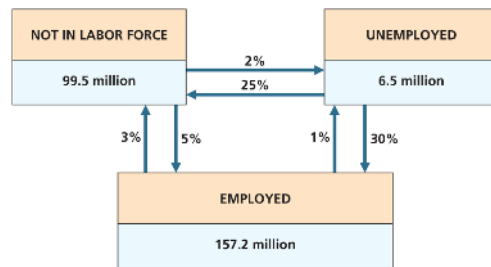
Category	Number (millions)	Share of labor force (percent)	Share of adult population (percent)
Employed workers	157.2	96.0	59.7 (employment ratio)
Unemployed workers	6.5	4.0 (unemployment rate)	2.5
Labor force (employed + unemployed workers)	163.7	100.0	62.2 (participation rate)
Not in labor force	99.5		37.8
Adult population (labor force + not in labor force)	263.2		100.0

Note: Figures may not add up because of rounding.
Source: The Employment Situation, January 2022, Table A-1.

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Changes in Employment Status in a Typical Month

| 3-5 · UNEMPLOYMENT



The arrows between two boxes represent changes from one employment status to another; the labels on the arrows show the fraction of the people in one status who switch to the other status in a typical month.

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Unemployment Duration

| 3-5 · UNEMPLOYMENT

- Most unemployment spells are of short duration
 - Unemployment spell = period of time an individual is continuously unemployed
 - Duration = length of unemployment spell
- Most unemployed people on a given date are experiencing unemployment spells of long duration

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Types of Unemployment

| 3-5 · UNEMPLOYMENT

- Frictional unemployment
 - Search activity of firms and workers due to heterogeneity
 - Matching process takes time
- Structural unemployment
 - Chronically unemployed: workers who are unemployed a large part of the time
 - The long-term and chronic unemployment that exists even when the economy is not in a recession

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The Job Market Is Hell

| 3-5 · UNEMPLOYMENT

By Annie Lowrey, *The Atlantic*, Sept 8



Young people are using ChatGPT to write their applications; HR is using AI to read them; no one is getting hired.

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Natural Rate of Unemployment

| 3-5 · UNEMPLOYMENT

- The natural rate of unemployment ($\bar{u}$)
 - when output and employment are at full-employment levels
 - = frictional + structural unemployment
- Cyclical unemployment: difference between actual unemployment rate and natural rate of unemployment ($u - \bar{u}$)

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Alternative Unemployment Measures

| 3-5 · UNEMPLOYMENT

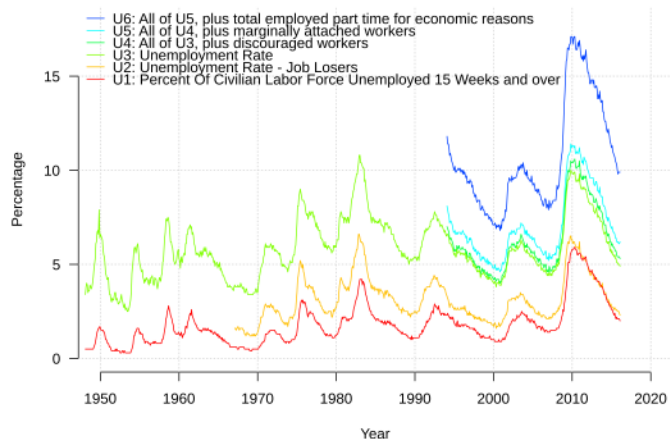
- U-1: unemployed 15 weeks or more
- U-2: counts job losers or persons who completed temporary jobs
- U-3: the official rate
- U-4: like U-3 but adds discouraged workers
- U-5: like U-4 but adds marginally attached workers
- U-6: like U-5 but adds persons who are employed part time for economic reasons

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Alternative Measures of Unemployment, 1994–2021

| 3-5 · UNEMPLOYMENT

6 MEASURES OF UNEMPLOYMENT



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- ✓ 3-1 · The Production Function
- ✓ 3-2 · The Demand for Labor
- ✓ 3-3 · The Supply of Labor
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- ✓ 3-5 · Unemployment
- ▶ 3-6 · Okun's Law

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Okun's Law (Potential GNP: Its Measurement and Significance. 1962)

| 3-6 - OKUN'S LAW

- Relationship between output (relative to full-employment output) and cyclical unemployment

$$\frac{\bar{Y} - Y}{\bar{Y}} = 2(u - \bar{u}) \quad (3.5)$$

- For every 1 percentage point increase in cyclical unemployment, output falls 2% below its potential.



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Okun's Law..

| 3-6 - OKUN'S LAW

A worked example. Suppose $\bar{u} = 4\%$ and full-employment output $\bar{Y} = \$20,000$ billion.

- Actual unemployment is $u = 5\% \rightarrow$ cyclical unemployment

$$u - \bar{u} = 5\% - 4\% = 1\%$$

- Okun's law: output gap is 2× cyclical unemployment

$$\frac{\bar{Y} - Y}{\bar{Y}} = 2(u - \bar{u}) = 2 \times 1\% = 2\%$$

- So actual output falls short of full employment by

$$\bar{Y} - Y = 2\% \times \$20,000 \text{ bn} = \$400 \text{ billion}$$

Takeaway: a 1 pp rise in unemployment costs about **\$400 billion** of output — a big number for a “small” gap.

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Okun's Law..

| 3-6 - OKUN'S LAW

- Why is the Okun's Law coefficient 2, and not 1?
 - Other things happen when cyclical unemployment rises: Labor force falls, hours of work per worker decline, average productivity of labor declines
 - Result is 2% reduction in output associated with 1 percentage point increase in unemployment rate
- Alternative formulation if average growth rate of full-employment output is 3%:

$$\Delta Y/Y = 3 - 2\Delta u \quad (3.6)$$

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When Is Cyclical Unemployment Zero?

| 3-6 - OKUN'S LAW

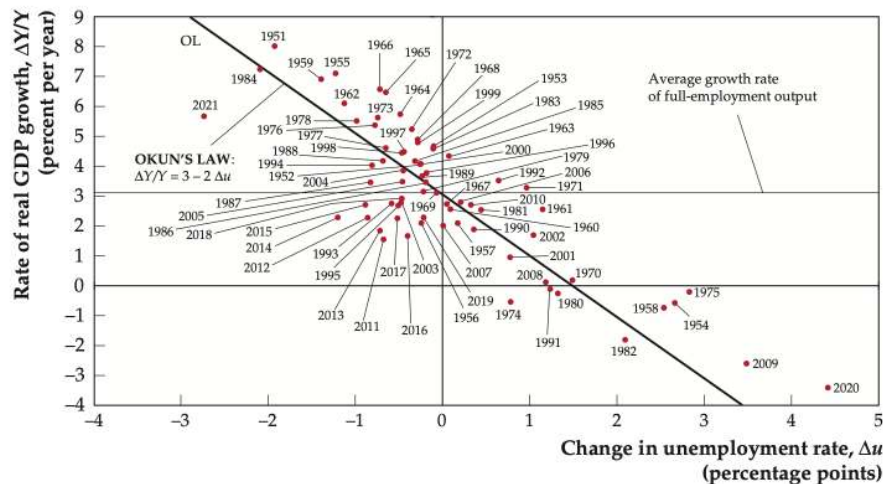
- Exactly when $Y = \bar{Y}$ (the economy is at full employment) — the full-employment condition itself, restated in unemployment terms
- Same idea as our very first question: at full employment $u = \bar{u}$, so cyclical unemployment = 0, while frictional + structural ($= \bar{u}$) still remain

State of economy	Output	Unemployment	Cyclical unemployment
Recession	$Y < \bar{Y}$	$u > \bar{u}$	Positive
Full employment	$Y = \bar{Y}$	$u = \bar{u}$	Zero

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Okun's Law in the United States: 1951–2021

| 3-6 - OKUN'S LAW



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► Numerical Problems

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U.S. Economy Data — Productivity and the MPN

| NUMERICAL PROBLEMS

The following data give real GDP, Y, capital, K, and labor, N, for the U.S. economy in various years

Var.	1960	1970	1980	1990	2000	2010	2020
Y	3,262	4,954	6,764	9,371	13,138	15,649	18,385
K	3,821	5,767	8,293	11,298	15,887	19,373	23,539
N	65	78	99	117	134	140	158

Units and sources are the same as in Table 3.1. Assume that the production function is $Y = AK^{0.3}N^{0.7}$

1. By what percentage did U.S. 'A' grow in each decade?

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U.S. Economy Data — Productivity and the MPN (continued)

| NUMERICAL PROBLEMS

Assume the production function is $Y = AK^{0.3}N^{0.7}$ (data on the previous slide).

2. What happened to the MPN between 1960 and 2020? Calculate the MP numerically as the extra output gained by adding 1 million workers in each of the two years. (The data for employment, N, are measured in millions of workers, so an increase of 1 million workers is an increase of 1.0.)

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Problem: RKn Widget, Inc.

| NUMERICAL PROBLEMS

RKn Widget, Inc. has the following production function.

Number of Workers	Number of Widgets Produced
0	0
1	8
2	15
3	21
4	26
5	30
6	33

a. Find the MPN for each level of employment.

b. RKn can get \$5 for each widget it produces. How many workers will it hire if the nominal wage is \$38? If it is \$27? If it is \$22?

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Problem: RKn Widget, Inc. (continued)

| NUMERICAL PROBLEMS

- c. Graph the relationship between RKn's labor demand and the nominal wage. How does this graph differ from a labor demand curve? Graph RKn's labor demand curve.
- d. With the nominal wage fixed at \$38, the price of widgets doubles from \$5 each to \$10 each. What happens to RKn's labor demand and production?
- e. With the nominal wage fixed at \$38 and the price of widgets fixed at \$5, the introduction of a new automatic widget maker doubles the number of widgets that the same number of workers can produce. What happens to labor demand and production?
- f. What is the relationship between your answers to part (d) and part (e)? Explain.

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Thank You

I will not be taking questions.!

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